

Concept Erasing in Large Language Models: Evaluating Machine Unlearning Techniques for Selective Knowledge Removal

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Abstract—Large Language Models (LLMs) can encode broad conceptual knowledge acquired from large-scale training data, motivating methods for selectively erasing targeted knowledge without retraining the entire model. Although concept-erasure methods are commonly evaluated on individual targets, the additional degradation that may arise when multiple broad concepts are erased jointly remains insufficiently characterized. This study investigates multi-concept interference through a controlled comparison of ELM, a fine-tuning-based concept-erasure method, and a simplified activation-level implementation of the feature/parameter-based PISCES approach. Experiments are conducted on Gemma-2-2B-it using Harry Potter and WMDP-Cyber as target concepts under single-concept and joint-erasure conditions. We introduce the Concept Interference Score (CIS) to quantify changes in erasure efficacy between single-concept and joint-erasure settings. ELM produces near-zero CIS values (-0.1% for Harry Potter and -0.8% for WMDP-Cyber) at the tested training scale, while PISCES exhibits substantial and concept-dependent interference under dual suppression, with CIS values of -18.7% for Harry Potter and -77.5% for WMDP-Cyber. The PISCES intervention-strength analysis further shows an erasure–utility trade-off across the tested suppression levels, while a separate higher-magnitude directional-steering configuration produces severe degradation in retained utility. Because ELM and PISCES use different efficacy measures, their absolute CIS magnitudes should not be interpreted as directly comparable effect sizes. Overall, the results show that multi-concept interference is measurable and concept-dependent under the tested PISCES configuration, whereas ELM exhibits negligible measured interference at the tested training scale.

Index Terms—Machine Unlearning, Large Language Models, Concept Erasure, Model Editing, Concept Interference Score

I. INTRODUCTION

Large Language Models (LLMs) are trained on massive, largely uncensored web-scale text corpora, and prior work has shown that such models can memorize and, under certain conditions, be induced to regurgitate portions of their training data [6]—raising concerns when that data includes sensitive dual-use information (e.g., biosecurity or cybersecurity knowledge) [3], copyrighted or licensed material, and other content that creates safety, legal, or ethical risk. This has motivated a growing body of work on machine unlearning for LLMs [8]:

techniques that attempt to selectively remove a targeted piece of knowledge from a trained model without retraining it from scratch and without destroying its general capabilities.

Within this space, concept erasure specifically targets broad conceptual knowledge (e.g., “how to build a bioweapon” or “Harry Potter lore” [7]) rather than individual training examples or isolated facts. Two recent and representative approaches are ELM (Erasing Conceptual Knowledge from Language Models) [1], which reformulates unlearning as a fine-tuning problem guided by the model’s own introspective classification ability, and PISCES (Precise In-Parameter Concept Erasure in Large Language Models) [2], which instead disentangles model parameters into interpretable features and suppresses the features associated with the target concept. These represent two fundamentally different design philosophies—fine-tuning-based versus feature/parameter-based erasure—that have been evaluated on different target concepts.

Both ELM [1] and PISCES [2] report their results almost exclusively for single-concept erasure. However, real-world deployment rarely has the luxury of a single sensitive concept. A production LLM intended for public use typically needs several sensitive domains removed at once—for example biosecurity, cybersecurity, and copyrighted content simultaneously—because leaving any one of them intact is already a deployment blocker.

This creates a specific and concrete dilemma: it is currently unknown whether erasing multiple concepts at the same time is simply the sum of erasing each concept individually, or whether the concepts interact with each other inside the model’s parameters in a way that makes joint erasure harder, less complete, or more damaging to general utility than erasing each concept alone would suggest.

This gap has not been closed for two related reasons. First, the evaluation protocols proposed alongside ELM [1] and PISCES [2] were designed to compare methods against each other on one concept at a time; none of these metrics is constructed to isolate or quantify the additional degradation that arises specifically from removing several concepts

together. Second, the two methods being compared operate on structurally different mechanisms (LoRA-based fine-tuning versus feature/parameter-based editing), and it is an open question whether that structural difference makes one approach more or less prone to concept interference.

Building directly on this gap, this study evaluates ELM [1] and PISCES [2] side-by-side under a multi-concept erasure setting—erasing Harry Potter and WMDP-Cyber jointly rather than individually, and comparing the outcome against erasing each concept alone using the same base model and evaluation protocol. To make that comparison precise, this study introduces the Concept Interference Score (CIS), specifically designed to quantify the extra degradation in erasure quality caused by joint erasure.

RQ1: Does jointly erasing multiple concepts from an LLM cause measurably greater degradation in forgetting quality, for each individual concept, than erasing that concept alone?

RQ2: Do fine-tuning-based (ELM) [1] and feature/parameter-based (PISCES) [2] erasure methods exhibit different levels of multi-concept interference under their respective experimental settings, as measured by method-specific CIS?

The aim of this study is to systematically evaluate ELM [1] and PISCES [2] under single- and multi-concept unlearning, to introduce and validate the Concept Interference Score (CIS), and to analyze which architectural paradigm better preserves both erasure effectiveness and general model utility.

II. RELATED WORK

Prior efforts to remove undesirable knowledge from LLMs fall into three related but distinct problem families: knowledge editing, which corrects or inserts discrete factual associations; representation-level unlearning, which suppresses a model’s ability to produce content from a targeted domain; and concept-level (feature-based) erasure, which targets broader conceptual knowledge rather than individual facts. ELM [1] and PISCES [2], the two methods directly examined in this study, belong to the latter category, but their relationship to the fact-editing and representation-unlearning literature helps clarify what remains unaddressed regarding multi-concept erasure.

MEMIT [4] and AlphaEdit [5] represent the knowledge-editing family. MEMIT updates a language model with many factual associations at once by computing a batched parameter update distributed across multiple MLP layers, and its original evaluation demonstrates that this batching scales to thousands of simultaneous fact edits. AlphaEdit extends this locate-then-edit paradigm by projecting each parameter update onto the null space of preserved-knowledge representations, and reports that this reduces disruption to unrelated facts under sequential editing. Both methods therefore already handle large numbers of simultaneous edits and both explicitly analyze degradation that arises as edits accumulate—but this degradation is disruption to unrelated, previously preserved facts caused by inserting or correcting many individual factual associations, not interference between multiple broad concepts that are each

being erased from the model. Neither paper’s evaluation is structured around erasing broad conceptual knowledge, and neither reports a metric that isolates degradation specifically attributable to jointly erasing multiple such concepts.

RMU [3], introduced alongside the WMDP benchmark, unlearns targeted knowledge by steering a model’s intermediate representations on forget-set inputs toward a random vector while preserving representations on retain-set inputs. WMDP itself spans biosecurity, cybersecurity, and chemical-security domains, and RMU’s original release includes a model unlearned across more than one of these domains rather than a single domain in isolation. This makes RMU/WMDP directly relevant to the WMDP-Cyber concept used in the present study. However, the original RMU/WMDP evaluation reports forgetting and utility-retention accuracy in aggregate across the domains it targets; it does not report a metric that isolates how much additional degradation, if any, arises specifically from unlearning multiple domains together relative to unlearning each alone. Multi-domain coverage in the training or evaluation set is therefore not equivalent to an explicit measurement of cross-concept interference.

ELM [1] and PISCES [2] are the two paradigms compared directly in this study. ELM reformulates concept erasure as a fine-tuning problem, using low-rank adapters guided by the model’s own introspective classification of concept-related content to reduce generation probabilities for a target concept while preserving general capability. PISCES instead operates by disentangling model parameters into interpretable features, identifying those associated with a target concept, and suppressing them directly in parameter space. This study’s own implementation of PISCES differs from this original mechanism; the distinction is described in Section V. Both ELM’s and PISCES’s original evaluations assess erasure efficacy, specificity, and (for PISCES) robustness on target concepts drawn from separate domains—biosecurity, cybersecurity, or literary content—but each concept is erased and evaluated in an independent run rather than jointly with another concept in a single intervention. Neither paper’s evaluation protocol is designed to measure whether erasing two concepts together degrades either concept’s erasure more than erasing it alone would.

Taken together, the surveyed methods each address a piece of the surrounding problem space without directly closing the gap this study targets. RMU’s original evaluation includes models unlearned on more than one WMDP domain, and MEMIT supports batched editing of many individual facts, but neither setting provides a metric that isolates the additional degradation specifically attributable to jointly erasing multiple broad concepts. ELM and PISCES likewise evaluate target concepts in independent erasure settings and do not introduce a metric designed to quantify cross-concept interference under a joint erasure intervention. This study addresses this narrower gap by jointly erasing Harry Potter and WMDP-Cyber within a single ELM or PISCES intervention and introducing the Concept Interference Score (CIS) to quantify the resulting degradation relative to single-concept erasure.

TABLE I
COMPARISON OF EXISTING CONCEPT ERASURE METHODS AND THE PROPOSED RESEARCH

Method	Erasure Mechanism	Selective	Multi-Concept	Scalability	Interference	New Metric	Utility Pres.	Robustness
RMU [3]	Representation-level	✓	Partial*	NR	×**	×	✓	✓
MEMIT [4]	Mass Parameter Editing	✓	Partial***	✓	×**	×	✓	NR
AlphaEdit [5]	Parameter Editing	✓	Partial***	✓	×**	×	✓	NR
ELM [1]	Fine-tuning (LoRA)	✓	×	×	×	×	✓	✓
PISCES [2]	Feature/Parameter Editing	✓	×	×	×	×	✓	✓
Proposed Comparative (ELM vs. PISCES)	Comparative	✓	✓	✓	✓	✓(CIS)	✓	✓

Legend: ✓ = Supported, × = Not Addressed, NR = Not Explicitly Reported.

* RMU’s original release includes a model unlearned across more than one WMDP domain, but this reflects multi-domain training/evaluation coverage, not an explicit measurement of cross-concept erasure interference as defined in this paper.

** Marked × for the CIS-style cross-concept interference construct specifically: these papers analyze disruption to unrelated preserved/retained knowledge under accumulating edits, which is a related but distinct form of degradation from the interference between jointly erased target concepts measured by CIS.

*** Supports large-scale batched editing of individual facts, not joint erasure of broad concepts.

III. DATASET DESCRIPTION

The following openly accessible datasets were used, consistent with the datasets already employed by ELM and PISCES.

A. Harry Potter Corpus

A Harry Potter text corpus was used as one of the target concepts for erasure. This dataset is openly available via the HuggingFace “harry_potter_tiny” collection and served as a well-understood, low-risk concept for validating the multi-concept erasure pipeline.

B. WMDP-Cyber Corpus

The WMDP-Cyber forget corpus was used as the second target concept, accessible via HuggingFace (cais/wmdp-corpora). WMDP-Cyber represents a genuinely sensitive, dual-use concept, allowing the joint-erasure experiments to reflect a realistic deployment scenario.

C. Retention / Utility Benchmarks

MMLU was used to measure retained general knowledge, and Forget/Retain Perplexity (PPL) was used to measure PISCES generation quality under intervention.

D. Note on WMDP-Bio

WMDP-Bio was excluded from primary experiments as its access is gated behind a request form due to biosecurity sensitivity.

IV. DATA PREPROCESSING

- Text segmentation: corpora segmented into fixed-length samples (≤ 700 characters) to create the erase-set.
- Sampling: a bounded number of samples ($n = 20$ per concept, due to memory constraints detailed in Section VII) drawn from each forget corpus.
- Conditional context prepending: concept-conditioning prefixes used for ELM’s conditional erasure formulation.
- Retain-set construction: a retention dataset of unrelated text assembled to measure specificity.
- Held-out evaluation split: each concept’s data split into training forget-set and a disjoint MCQ evaluation set.
- Tokenization: all text tokenized using the base model’s native tokenizer (padding_side=‘left’, pad_token=eos_token).

TABLE II
EVALUATION METRICS USED IN THIS STUDY

Metric	Purpose and Measurement
Efficacy (Innocence)	Measures target concept erasure. For ELM, via MCQ accuracy on concept questions (%), where lower accuracy indicates stronger erasure. For PISCES, via forget-target perplexity (PPL), where higher PPL indicates stronger erasure.
Specificity	Evaluates preservation of unrelated knowledge via MMLU accuracy (ELM) or Retain-domain PPL (PISCES).
CIS (Proposed)	Quantifies degradation from joint erasure versus single-concept erasure, computed per-method using each method’s respective efficacy measure.

V. METHODOLOGY

This study follows a controlled comparative experimental design to evaluate ELM [1] and PISCES [2] under single-concept and multi-concept erasure settings on Gemma-2-2B-it. Fig. 1 illustrates the overall workflow of the evaluation framework, and Table II defines the evaluation metrics used in this study, including the proposed CIS.

Coherence and robustness, as general dimensions of erasure quality, are noted for completeness but were not experimentally measured in this study; reported results are limited to efficacy, specificity, and CIS as defined above.

A controlled, within-model comparative experiment was conducted on Gemma-2-2B-it. For ELM, four erasure conditions were evaluated: (A) Harry Potter only, (B) WMDP-Cyber only, (C) Harry Potter + WMDP-Cyber jointly, and (D) an unedited baseline. For PISCES, a single-target vs. dual-target design was used: a full alpha-suppression sweep ($\alpha \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$) was run for Harry Potter alone, WMDP-Cyber was suppressed alone at $\alpha = 0.0$, and both concepts were suppressed jointly in a single dual-target intervention at $\alpha = 0.0$ only; a joint alpha sweep was not performed. A separate, higher-magnitude directional concept-vector steering condition ($\gamma = 2.5$) was additionally evaluated as a distinct configuration, not as a point on the alpha-sweep.

Multi-Concept Erasure Evaluation Pipeline

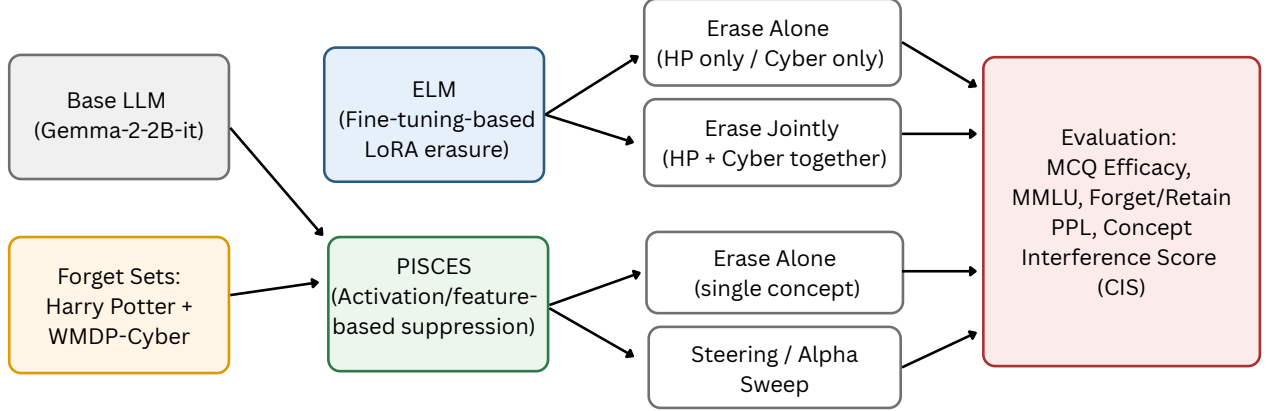


Fig. 1. Proposed methodology for comparative evaluation of multi-concept erasure using ELM and PISCES.

Base model: Gemma-2-2B-it, selected for its accessibility and suitability for activation-level intervention at moderate scale. Target concepts: Harry Potter and WMDP-Cyber, both openly accessible. WMDP-Bio was excluded due to gated-access requirements.

The evaluation protocol follows Table II. For ELM, efficacy was measured via MCQ accuracy on held-out concept questions, and specificity via MMLU accuracy. For PISCES, efficacy was measured via forget-target perplexity and specificity via Retain-domain perplexity. SHAP-based token attribution was additionally applied to the extracted Harry Potter concept direction from this study’s PISCES implementation, to examine its lexical localization.

ELM: A LoRA adapter (rank=4, alpha=4, dropout=0.05) was trained on layers 4–7 using a combined erasure, retention, and consistence loss ($\eta = 15, 20$ training samples per concept). For joint erasure, samples from both concepts were included within a single training run. This configuration reflects the final settings used to produce the reported results; no formal sweep over η was conducted.

PISCES: This study implements a simplified version of the PISCES mechanism and does not use the original method’s dictionary-learned sparse-autoencoder (SAE) feature extraction. Concept directions were instead approximated as the mean hidden-state activation difference between 20 forget-set samples and 20 retain-set samples per concept, pooled and normalized. Interventions were applied at Layer 12, selected according to the interpretability heuristic described in

Section VI. Two intervention regimes were tested: (i) scaled suppression along the concept direction at suppression level α , applied to the single-concept direction across the full sweep $\alpha \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ for Harry Potter, and at $\alpha = 0.0$ only for WMDP-Cyber and for the dual-concept (joint) condition; and (ii) directional concept-vector steering, $h' = h - \gamma v_{\text{concept}}$, at $\gamma = 2.5$, evaluated as a separate, higher-magnitude configuration.

The primary analysis compares alone vs. joint erasure results for each method and computes the corresponding CIS (RQ1). For ELM,

$$\text{CIS}_{\text{ELM}} = \frac{\text{Efficacy}_{\text{alone}} - \text{Efficacy}_{\text{joint}}}{\text{Efficacy}_{\text{alone}}}. \quad (1)$$

For PISCES, because efficacy is measured via forget-target PPL, where higher values indicate stronger erasure,

$$\text{CIS}_{\text{PISCES}} = \frac{\text{PPL}_{\text{joint}} - \text{PPL}_{\text{alone}}}{\text{PPL}_{\text{alone}}}, \quad (2)$$

so that a negative value consistently indicates weaker erasure under joint suppression for both methods. The secondary analysis compares the CIS values of ELM and PISCES to assess differences in multi-concept interference (RQ2). The available intervention-strength results, including the PISCES alpha sweep and directional-steering condition, are analyzed separately as supplementary evidence rather than as a symmetric cross-method intervention-strength comparison.

TABLE III
ELM RESULTS ACROSS ALL RUNS

Run	MMLU	Bio Acc.	Cyber Acc.
Baseline	0.5691	—	—
HP alone	0.5763	—	—
Cyber alone	0.5728	0.6489	0.3699
Joint	0.5728	0.6473	0.3729

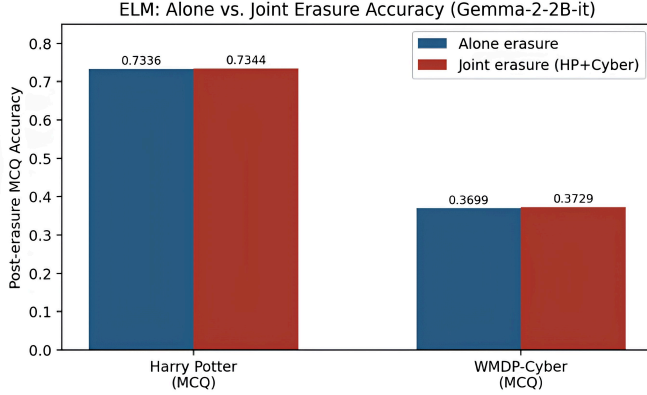


Fig. 2. ELM alone vs. joint post-erasure MCQ accuracy.

TABLE IV
ELM CONCEPT INTERFERENCE SCORE (CIS)

Concept	Alone	Joint	CIS
Harry Potter	0.7336	0.7344	−0.1%
WMDP-Cyber	0.3699	0.3729	−0.8%

VI. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experiments were run on Gemma-2-2B-it using a single Tesla T4 GPU (16GB VRAM) on Kaggle Notebooks. ELM used float32 training precision; post-hoc inference comparisons used float16. PISCES interventions were applied at Layer 12 of the base model.

B. ELM: Single-Concept and Multi-Concept Results

Table III reports MMLU and WMDP accuracy across all completed ELM runs. General utility (MMLU) remained stable (≈ 0.57 – 0.576) across the baseline, single-concept, and joint-erasure conditions.

Table IV reports the multi-concept Concept Interference Score (CIS), computed as

$$\text{CIS} = \frac{\text{Efficacy}_{\text{alone}} - \text{Efficacy}_{\text{joint}}}{\text{Efficacy}_{\text{alone}}} \quad (3)$$

using MCQ accuracy as the efficacy measure.

Both concepts show small negative CIS values (-0.1% for Harry Potter, -0.8% for WMDP-Cyber), indicating negligible measurable interference between concepts under joint erasure at this configuration ($n = 20$ samples, $\eta = 15$). Token-level

TABLE V
PISCES ALPHA SWEEP RESULTS (LAYER 12)

α	HP Forget PPL	Retain PPL
1.0 (none)	61.94	72.39
0.8	76.10	67.14
0.6	113.63	64.07
0.4	231.81	64.36
0.2	591.37	70.18
0.0 (full)	1668.10	86.97

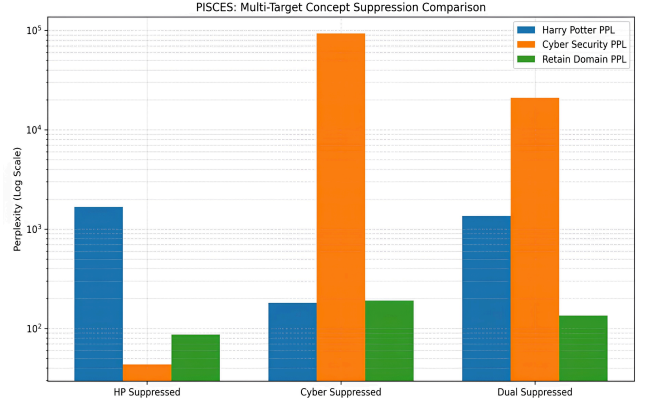


Fig. 3. PISCES Forget (Harry Potter) vs. Retain perplexity across the full alpha-suppression sweep (log scale).

probability analysis corroborates this: baseline vs. post-erasure next-token probabilities shifted by only approximately 0.7% (Harry Potter) and 0.1% (WMDP-Cyber), suggesting ELM’s absolute erasure effect at this sample size is itself weak—a finding discussed further in the limitations.

C. PISCES: Alpha Suppression Sweep

Table V reports Harry Potter forget-target perplexity (PPL) and Retain-domain PPL across a full alpha-suppression sweep at Layer 12 ($\alpha = 1.0$ denotes no intervention; $\alpha = 0.0$ denotes maximal suppression of the concept direction).

Forget PPL rises smoothly and monotonically as alpha decreases from 1.0 to 0.0 ($61.94 \rightarrow 1668.10$, a approximately $27\times$ increase), indicating progressively stronger erasure of the target concept. Retain PPL remains comparatively contained across the sweep, ranging from 64.07 (at $\alpha = 0.4$ – 0.6) to 86.97 (at $\alpha = 0.0$), i.e., at most a approximately $1.2\times$ increase over the 72.39 baseline. This is a substantially more graceful and controllable erasure–utility trade-off than the large-magnitude directional steering result reported later, indicating that scaled activation suppression (bounded to the $[0,1]$ range) is a materially more stable PISCES configuration than unconstrained directional-vector subtraction.

D. PISCES: Single-Target vs. Dual-Target (Multi-Concept) Suppression

To directly test multi-concept interference under PISCES, both Harry Potter and WMDP-Cyber were suppressed indi-

TABLE VI
PISCES SINGLE-TARGET VS. DUAL-TARGET SUPPRESSION

Condition	HP PPL	Cyber PPL	Retain PPL
Baseline	61.94	10.56	72.39
HP suppressed (alone)	1668.10	43.31	86.97
Cyber suppressed (alone)	180.82	93076.45	191.07
Dual (HP+Cyber jointly)	1356.47	20921.43	134.61

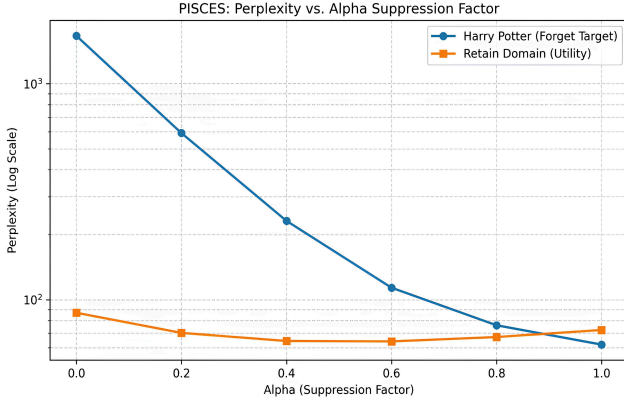


Fig. 4. PISCES multi-target concept suppression comparison across HP-only, Cyber-only, and dual (HP+Cyber) suppression conditions (log scale).

vidually (at $\alpha = 0.0$) and then jointly in a single dual-target intervention. Table VI reports forget-target PPL for both concepts and Retain-domain PPL under each condition.

Retain-domain PPL stays within a moderate range across all three suppression conditions (86.97–191.07, i.e., 1.2–2.6 $\times$ the 72.39 baseline), confirming that PISCES’s alpha-suppression mechanism preserves general utility far better than the large-magnitude gamma-steering configuration.

However, forget-target PPL tells a more nuanced story: in the dual-suppression condition, WMDP-Cyber forget-PPL falls sharply from 93,076.45 (suppressed alone) to 20,921.43 (suppressed jointly with Harry Potter)—a approximately 4.4 $\times$ reduction in erasure strength. Harry Potter forget-PPL also decreases, from 1668.10 (alone) to 1356.47 (jointly), a smaller but still measurable approximately 19% reduction. This constitutes direct empirical evidence of concept interference: suppressing two concepts simultaneously with PISCES measurably weakens the erasure of each, particularly the Cyber concept.

E. PISCES Concept Interference Score (CIS)

Using forget-target PPL as the efficacy measure (higher PPL indicates stronger erasure, the inverse orientation of the MCQ-accuracy-based CIS used for ELM), CIS is computed as

$$\text{CIS} = \frac{\text{PPL}_{\text{joint}} - \text{PPL}_{\text{alone}}}{\text{PPL}_{\text{alone}}}, \quad (4)$$

so that, consistent with the ELM convention, a negative value indicates weaker (worse) erasure under joint suppression.

TABLE VII
PISCES CONCEPT INTERFERENCE SCORE (CIS)

Concept	Alone / Joint Forget PPL	CIS
Harry Potter	1668.10 / 1356.47	−18.7%
WMDP-Cyber	93076.45 / 20921.43	−77.5%

TABLE VIII
PISCES DIRECTIONAL STEERING RESULTS ($\gamma = 2.5$)

Metric	Baseline	$\gamma = 2.5$
Forget PPL	58.50	302.63
Retain PPL	18.76	379,436.38

Both concepts show substantial negative CIS under PISCES—18.7% for Harry Potter and −77.5% for WMDP-Cyber—in contrast to ELM’s near-zero CIS (−0.1% and −0.8%, respectively). These results support RQ1 for PISCES under the tested dual-suppression configuration, while ELM shows only negligible measured interference at the tested training scale.

WMDP-Cyber shows substantially greater degradation than Harry Potter under the dual-suppression condition, indicating that the degree of interference is concept-dependent under the tested configuration.

F. PISCES: Directional Concept-Vector Steering (High-Magnitude Regime)

For comparison against the bounded alpha-suppression regime, a separate, higher-magnitude intervention was also tested: directional concept-vector steering,

$$h' = h - \gamma v_{\text{concept}}, \quad (5)$$

at $\gamma = 2.5$, Layer 12. This uses a different baseline evaluation pass than the alpha-sweep runs and should be read as a distinct, more aggressive PISCES configuration rather than a point on the same alpha-sweep curve.

Steering at $\gamma = 2.5$ substantially disrupts the target concept (Forget PPL: 58.50 $\rightarrow$ 302.63) but causes catastrophic degradation in retained capability (Retain PPL: 18.76 $\rightarrow$ 379,436.38—a $> 20,000\times$ increase), consistent with classic over-steering / model breakdown.

Read alongside the alpha-suppression results, this indicates that PISCES’s stability is highly configuration-dependent: the bounded [0,1] alpha-suppression mechanism achieves strong, controllable erasure with moderate utility cost, whereas unconstrained directional-vector subtraction at even a modest magnitude ($\gamma = 2.5$) collapses general capability entirely. Practitioners applying PISCES-style interventions should therefore prefer scaled, bounded suppression over large-magnitude directional steering.

G. SHAP Interpretability Analysis

To examine which tokens the extracted Harry Potter concept direction (v_{hp}) is most sensitive to, SHAP (SHapley Additive

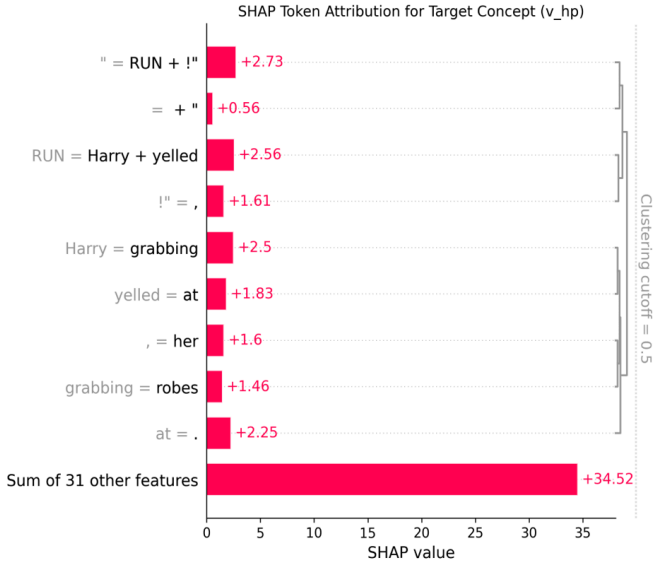


Fig. 5. SHAP token-attribution clustering for the Harry Potter concept direction (v_{hp}); bars show additive SHAP value contributions of token clusters to the concept-presence score.

exPlanations) was used to attribute a concept-presence score to input tokens for a representative Harry Potter passage. Fig. 5 shows the resulting token-clustering bar plot.

The highest-attribution clusters correspond to domain-entity and narrative-action tokens directly tied to the Harry Potter setting (e.g., “Harry”, “yelled”, “grabbing”, “robes”) rather than generic function words, suggesting that v_{hp} is substantially localized to concept-specific lexical content rather than broad syntactic structure.

This is consistent with—though not sufficient on its own to fully explain—the relatively graceful erasure/utility trade-off observed for alpha-suppression: a more lexically localized concept direction plausibly permits partial suppression with comparatively less disruption to unrelated generation.

A companion text-level SHAP visualization was also generated but is omitted from this manuscript: the underlying `shap.plots.text()` call renders as an interactive HTML widget rather than a static matplotlib figure, and the corresponding exported image file did not capture any content. Producing a static, paper-ready version of this visualization is left as a minor implementation item for future revisions.

H. Comparative Summary: ELM vs. PISCES

Table IX summarizes the trade-off across mechanism, compute, erasure efficacy, utility preservation, multi-concept interference, and stability.

Overall, ELM and PISCES reveal two distinct facets of the multi-concept erasure problem. ELM’s gradient-based fine-tuning, regularized by an explicit retention loss, yields highly stable utility preservation and negligible measured interference between concepts—but this near-zero CIS is entangled with a floor effect, since ELM’s absolute single-concept erasure was itself weak at the tested training scale ($n = 20$).

PISCES, by contrast, shows stronger single-concept erasure and substantial, asymmetric multi-concept interference under dual suppression, with CIS values of -18.7% for Harry Potter and -77.5% for WMDP-Cyber. At the same time, PISCES’s stability is highly configuration-dependent: the bounded alpha-suppression regime preserves utility reasonably well, while unconstrained directional-vector steering at even modest magnitude causes catastrophic collapse.

Taken together, these results suggest that meaningful empirical study of concept interference (RQ1) requires a method whose single-concept erasure is strong enough to produce a measurable signal—a condition PISCES’s alpha-suppression regime satisfies more clearly than ELM did in this compute-constrained setting (RQ2).

VII. LIMITATIONS AND FUTURE DIRECTIONS

- **Compute-constrained ELM training scale:** ELM results reflect a single, small training-sample regime ($n = 20$) due to a GPU memory constraint (Tesla T4, 16GB) that prevented larger-scale training; absolute ELM erasure conclusions should be scoped to this regime, and results may differ substantially at production scale (hundreds to thousands of samples, as used in the original ELM paper).
- **Possible floor effect in ELM CIS:** because ELM’s single-concept erasure was itself weak at $n = 20$, the near-zero ELM multi-concept CIS may partly reflect insufficient “erasure signal” for concepts to meaningfully interfere, rather than genuine architectural non-interference at scale. The PISCES results, where strong single-concept erasure co-occurs with substantial CIS, are consistent with this interpretation and motivate scaling up ELM training in future work to test whether its CIS remains near zero once its own single-concept erasure is strengthened.
- **Non-comparable CIS scales and sign conventions:** ELM CIS is accuracy-based (bounded, lower-is-better efficacy) while PISCES CIS is PPL-based (unbounded, higher-is-better efficacy); the two were computed with matched sign conventions in this study (negative = worse joint erasure) for qualitative comparability, but their absolute magnitudes are not on a common numeric scale. A normalized or log-scaled CIS variant, applicable across efficacy-metric types, is recommended future work for direct cross-method magnitude comparison.
- **Two non-identical PISCES baselines:** the alpha-suppression sweep/dual-suppression experiments (baseline Forget PPL=61.94, Retain PPL=72.39) and the directional gamma-steering experiment (baseline Forget PPL=58.50, Retain PPL=18.76) were run as separate

TABLE IX
COMPARATIVE TRADE-OFF SUMMARY

Dimension	ELM	PISCES
Mechanism	LoRA fine-tuning (layers 4–7), gradient-based	Direct activation intervention (Layer 12), training-free (alpha-suppression and directional steering variants)
Compute Erasure Efficacy	Moderate—training + evaluation passes Weak at $n = 20$ (floor effect); near-zero CIS (−0.1% to −0.8%)	Low per-run—single forward pass with hook Strong forget-PPL increase, especially for Cyber (up to approximately $8800\times$ baseline)
Utility Preservation	Strong—MMLU stable (~ 0.57) across all runs	Moderate under alpha-suppression (Retain PPL $1.2\text{--}2.6\times$ baseline); severe collapse under large-magnitude gamma-steering ($\gamma = 2.5$, Retain PPL $> 20,000\times$ baseline)
Multi-Concept Interference (CIS)	Near-zero (−0.1% HP, −0.8% Cyber)—concepts largely independent, but likely a floor effect given weak single-concept erasure	Substantial and asymmetric (−18.7% HP, −77.5% Cyber)—joint suppression measurably weakens erasure, especially for the second concept
Stability	No formal intervention-strength sweep; final results use $\eta = 15$.	Alpha-suppression shows a smooth erasure–utility trade-off; $\gamma = 2.5$ causes severe utility degradation

evaluation passes with slightly different baseline values, likely reflecting minor differences in evaluation batching or sample ordering. Results within each experiment are internally consistent, but absolute PPL values should not be compared across the two experiments without re-running both under an identical evaluation harness.

- **Incomplete SHAP interpretability output:** the text-level SHAP visualization (intended as a companion to Fig. 5) could not be exported as a static image, since `shap.plots.text()` produces an interactive HTML widget rather than a matplotlib figure; only the token-clustering bar plot (Fig. 5) is reported. Producing a publication-ready static rendering of the text-level view is left as future work.
- **WMDP-Bio exclusion:** WMDP-Bio was excluded due to gated dataset access; extending this study to a third, higher-sensitivity concept would strengthen generalizability claims, particularly given the asymmetric interference observed between Harry Potter and WMDP-Cyber under PISCES.
- **Single base model:** all experiments used Gemma-2-2B-it; replication on larger models (e.g., Llama-3.1-8B-it) would clarify whether the observed ELM/PISCES contrast in interference behavior generalizes across model scale.
- **Hyperparameter search granularity:** the PISCES alpha sweep (six points, $\alpha \in [0, 1]$) and single steering magnitude ($\gamma = 2.5$) still represent a relatively coarse grid; a finer sweep, and a corresponding alpha-based dual-suppression sweep (rather than a single $\alpha = 0.0$ dual-suppression point), would allow a full erasure–utility–interference frontier to be characterized for PISCES.

For RQ2, the two methods exhibit different multi-concept interference behavior under their respective experimental settings. ELM shows near-zero measured CIS for both target

concepts, whereas PISCES shows substantial negative CIS under dual suppression. These results indicate a qualitative difference in the observed interference behavior of the two methods.

Because ELM and PISCES use different efficacy measures—MCQ accuracy for ELM and forget-target PPL for PISCES—the absolute CIS magnitudes should not be interpreted as directly comparable effect sizes.

VIII. CONCLUSION

This study presented a controlled, side-by-side evaluation of two structurally different LLM concept-erasure approaches—ELM (fine-tuning-based) and this study’s simplified activation-level implementation of PISCES (feature/parameter-based)—on Gemma-2-2B-it under single-concept and multi-concept erasure conditions, and introduced the Concept Interference Score (CIS) to quantify degradation caused specifically by joint erasure.

Under the tested training scale of 20 samples per concept, ELM produced near-zero multi-concept CIS values (−0.1% for Harry Potter and −0.8% for WMDP-Cyber), while the reported MMLU results across the baseline, single-concept, and joint-erasure conditions remained within a narrow range (approximately 0.57–0.576).

PISCES, evaluated under the tested suppression configurations, showed substantial, concept-dependent multi-concept interference: dual-target suppression weakened erasure of both concepts relative to single-concept suppression, with WMDP-Cyber showing greater degradation than Harry Potter under the tested dual-suppression condition (CIS of −18.7% for Harry Potter and −77.5% for WMDP-Cyber).

A separate, higher-magnitude directional-steering configuration of PISCES ($\gamma = 2.5$) produced severe degradation in retained utility under the tested setting.

Because ELM and PISCES use different efficacy measures—MCQ accuracy for ELM and forget-target

perplexity for PISCES—their CIS values are computed on different scales, and the absolute magnitudes of the reported CIS values should not be interpreted as directly comparable effect sizes across methods.

Within this scope, the results indicate that multi-concept interference, as measured by CIS, was substantial and concept-dependent for PISCES under the tested joint-erasure setting, whereas ELM showed negligible measured interference at the tested training scale.

These findings motivate future work on (i) evaluating ELM at larger training scales, (ii) developing a more cross-method-comparable interference metric, and (iii) extending PISCES intervention-strength evaluation beyond the tested configurations.

REFERENCES

- [1] R. Gandikota et al., “Erasing Conceptual Knowledge from Language Models,” in *Advances in Neural Information Processing Systems* (*NeurIPS*), 2025.
- [2] Y. Gur-Arieh et al., “Precise In-Parameter Concept Erasure in Large Language Models,” in *Proc. 2025 Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 18986–19006, 2025.
- [3] N. Li et al., “The WMDP Benchmark: Measuring and Reducing Malicious Use With Unlearning,” in *Proc. Int. Conf. on Machine Learning (ICML)*, 2024.
- [4] K. Meng, A. S. Sharma, A. Andonian, Y. Belinkov, and D. Bau, “Mass-Editing Memory in a Transformer,” in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2023.
- [5] J. Fang et al., “AlphaEdit: Null-Space Constrained Knowledge Editing for Language Models,” in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2025.
- [6] N. Carlini, F. Tramèr, E. Wallace, M. Jagielski, A. Herbert-Voss, K. Lee, A. Roberts, T. Brown, D. Song, U. Erlingsson, O. Oprea, and C. Raffel, “Extracting Training Data from Large Language Models,” in *Proc. 30th USENIX Security Symposium (USENIX Security 21)*, pp. 2633–2650, 2021.
- [7] R. Eldan and M. Russinovich, “Who’s Harry Potter? Approximate Unlearning in LLMs,” arXiv preprint arXiv:2310.02238, 2023.
- [8] Y. Yao, X. Xu, and Y. Liu, “Large Language Model Unlearning,” arXiv preprint arXiv:2310.10683, 2023.